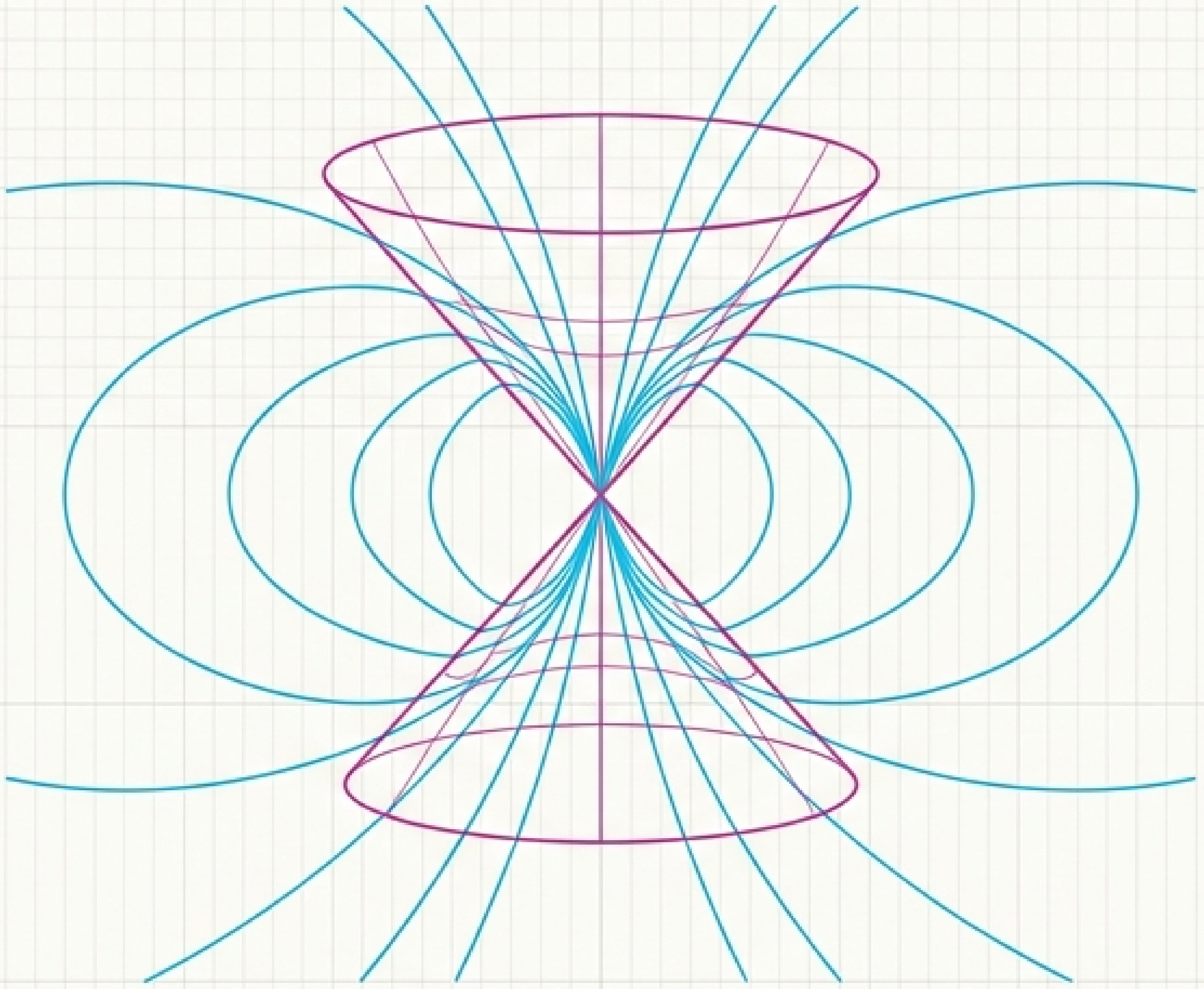


Foundations of Modern Physics

Translating the mathematical architecture of Relativity and Electrostatics into visual geometry.

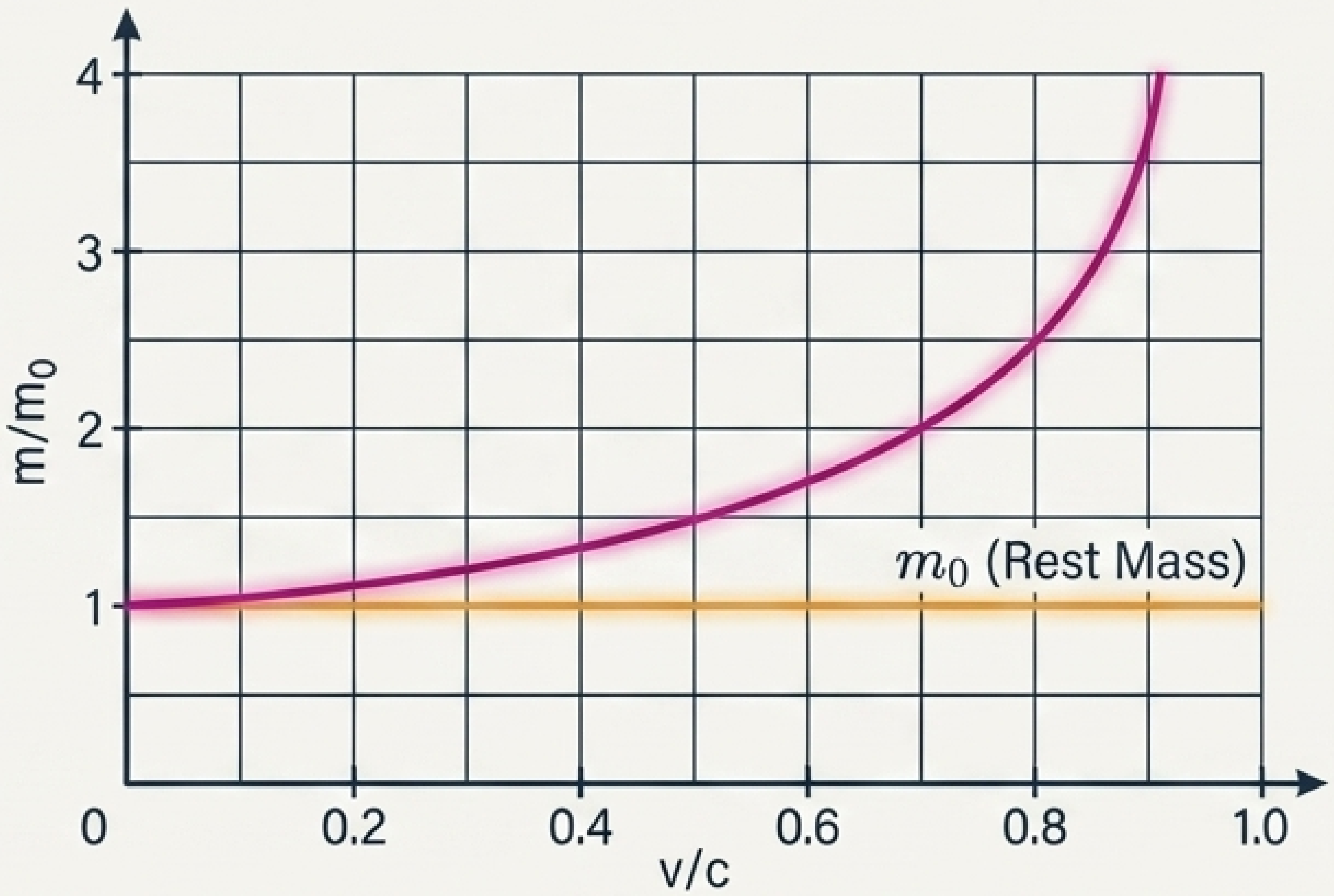
The Relativistic Shift

As the velocity approaches the speed of light, classical intuition breaks down, Ithe relativistic down to reminitize the mass, instory from mass foured in the relativistic mass. *The* are that is.

Lorentz factor: $\gamma = \dfrac{1}{\sqrt{1-\beta^2}}$

$$\beta = v/c$$

$$m = \frac{m_0}{\sqrt{1-\beta^2}}$$



Key Takeaway: Mass has relative meaning. It is not a fixed property of an object, but a dynamic response to motion.

Relativistic Mechanics: Newton's 2nd Law Revised

$$\boxed{p = mv}$$

Step 2

$$p = \frac{m_0 v}{\sqrt{1-\beta^2}}$$

Step 3

$$\boxed{F = \frac{dp}{dt}}$$

Step 4

$$F = \frac{d}{dt}\left(\frac{m_0 v}{\sqrt{1-\beta^2}}\right)$$

Only when $v \ll c$

$$F = m_0 a$$

NotebookLM

The Kinetic Energy Theorem

The Setup

$$E_k = \int F\,dx = \int v\,dp$$

The Integration

Integration by Parts

$$d(pv) = p\,dv + v\,dp$$

The Resolution

Energy of Motion

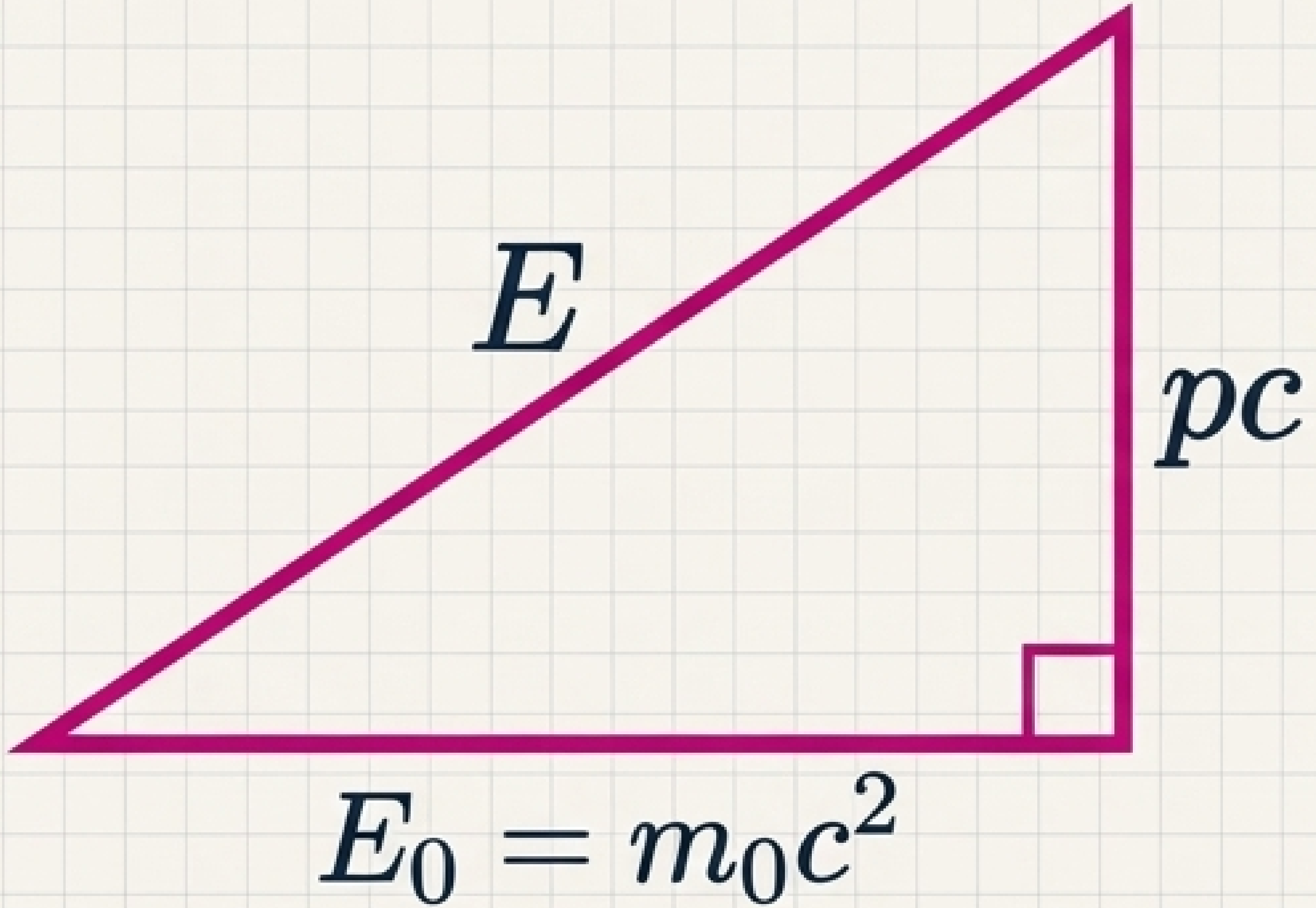
$$E_k = mc^2 - m_0c^2$$

Total Energy (Moving)

Rest Energy

Mass-Energy Equivalence & The Photon

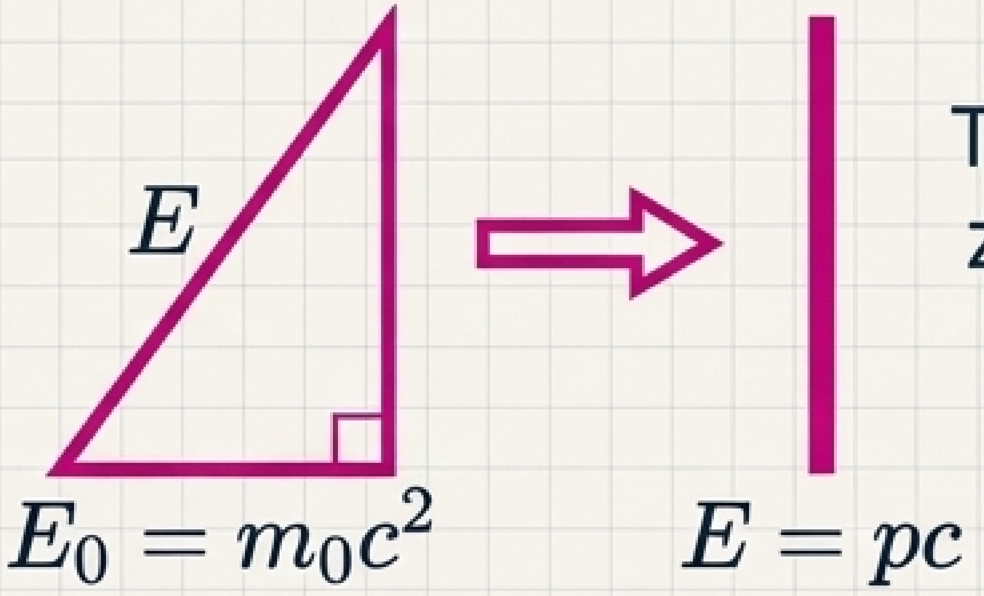
The Energy-Momentum Triangle



$$\boxed{E^2 = E_0^2 + (pc)^2}$$

The Photon Exception

If moving objects gain mass, how does light travel at c?



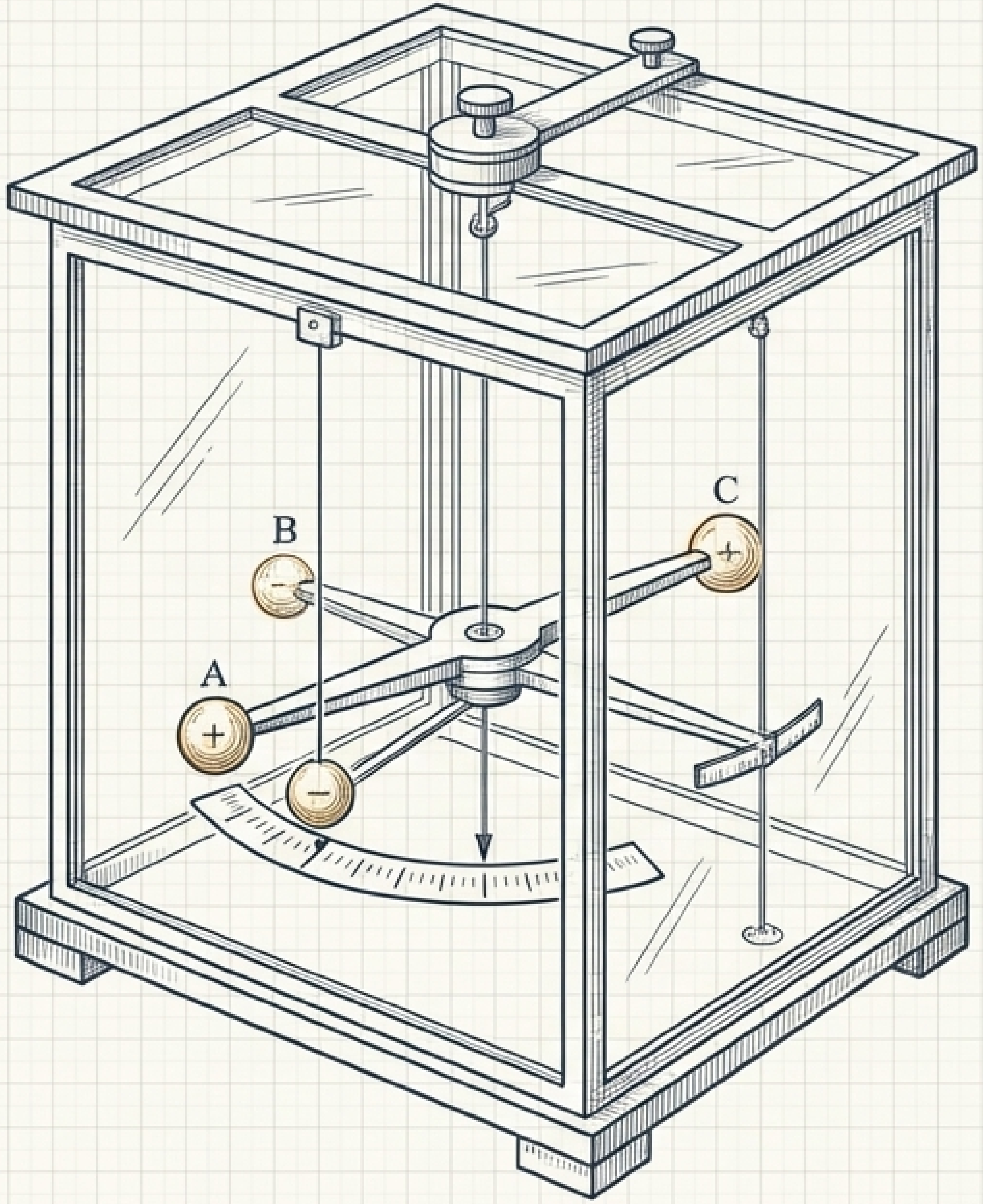
The photon has zero rest mass: $m_0 = 0$

$$E = h\nu \qquad p = \frac{h}{\lambda}$$

The Nuclear Engine: Transforming Mass Defect

$$\Delta E = (\Delta m)c^2$$

Heavy Fission (Uranium-235)	Light Fusion (Deuterium)
${}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{139}_{54}Xe + {}^{95}_{38}Sr + 2{}^{1}_{0}n$	${}^{2}_{1}H + {}^{2}_{1}H \rightarrow {}^{4}_{2}He$
Mass Defect: $\Delta m = 0.22u$	Mass Defect: $\Delta m = 0.026u$
Energy Yield: $E \approx 200\ \mathrm{MeV}$	Energy Yield: $E \approx 24\ \mathrm{MeV}$
1g U-235 yields: $Q = 8.5 \times 10^{10}\ \mathrm{J}$	Extreme Condition Requirement: Requires $10^8\ \mathrm{K}$ temperature to overcome Coulomb repulsion.



The Electrostatic World

From discrete, invisible charges to continuous fields of force.

Core Principle

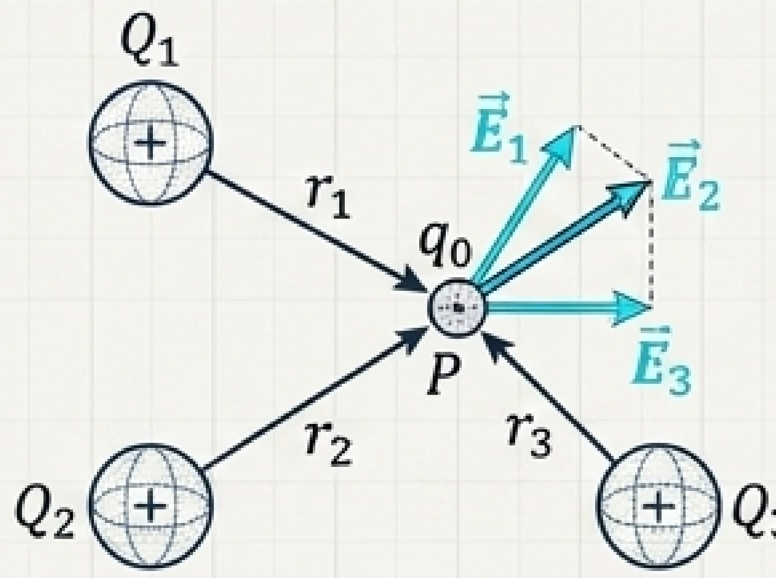
$$q = \pm ne \quad (n = 1, 2, 3, \ldots)$$

The **total electric charge** of an isolated system is absolutely conserved. (Law of Conservation of Charge)

The Charge Distribution Topology Matrix

Point Charges (Σ)

$$\boldsymbol{E} = \frac{1}{4\pi\varepsilon_0} \frac{Q}{r^2} \hat{\boldsymbol{e}}_r$$



$$\boldsymbol{E}_{total} = \sum_i \boldsymbol{E}_i = \frac{1}{4\pi\varepsilon_0} \sum_i \frac{Q_i}{r_i^2} \hat{\boldsymbol{e}}_i$$

Continuous Distributions (∫)

$$d\boldsymbol{E} = \frac{1}{4\pi\varepsilon_0} \frac{dq}{r^2} \hat{\boldsymbol{e}}_r$$

Volume (ρ)

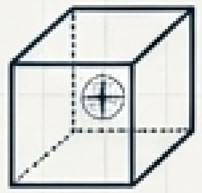
$dq = \rho dV$

Surface (σ)

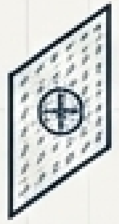
$dq = \sigma dS$

Line (λ)

$dq = \lambda dl$



$$\boldsymbol{E} = \int \frac{1}{4\pi\varepsilon_0} \frac{\rho \hat{\boldsymbol{e}}_r}{r^2} dV$$



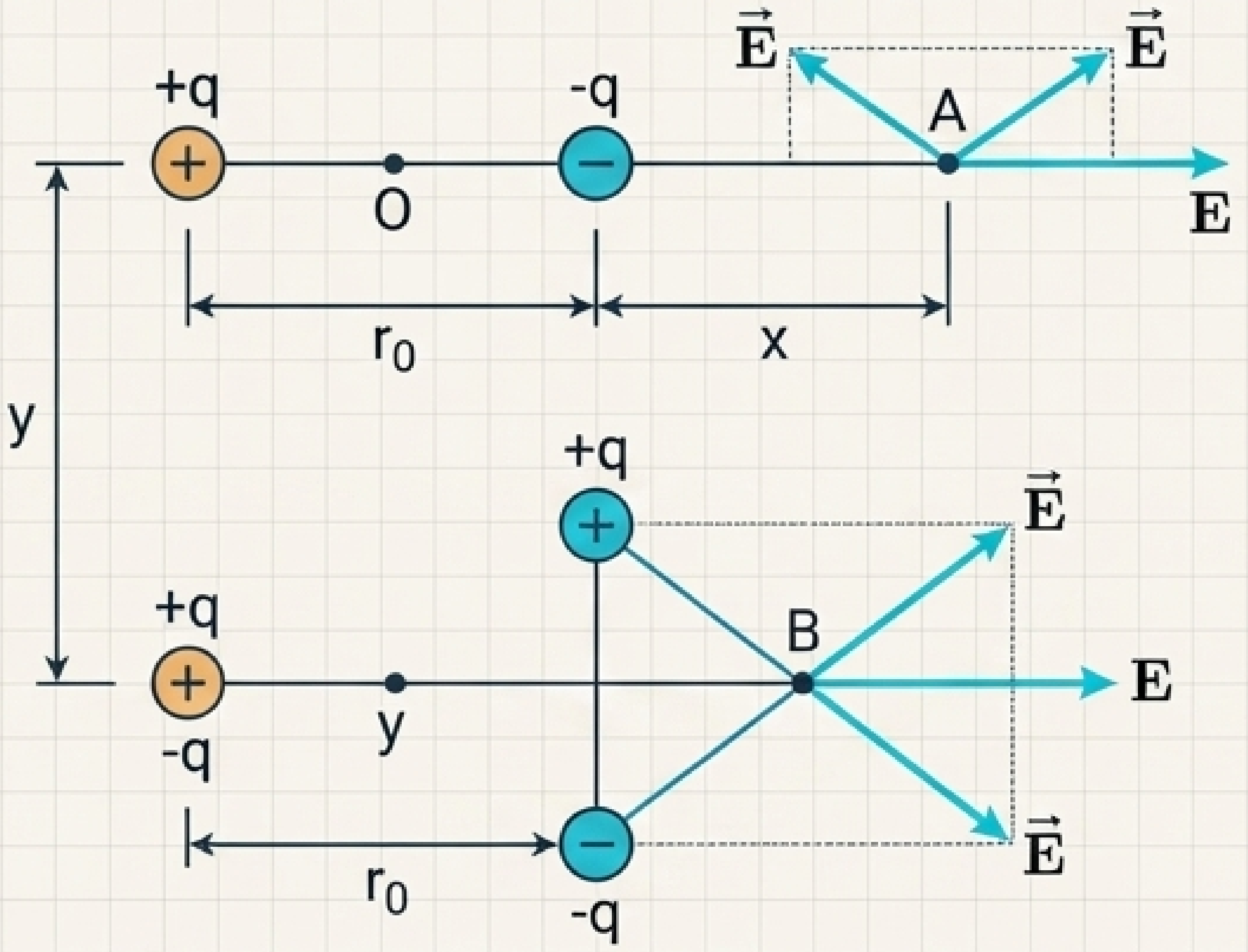
$$\boldsymbol{E} = \int \frac{1}{4\pi\varepsilon_0} \frac{\sigma \hat{\boldsymbol{e}}_r}{r^2} dS$$

$$\boldsymbol{E} = \int \frac{1}{4\pi\varepsilon_0} \frac{\lambda \hat{\boldsymbol{e}}_r}{r^2} dl$$

Insight: The fundamental force formula remains constant; only the geometric definition of the source (dq) adapts.

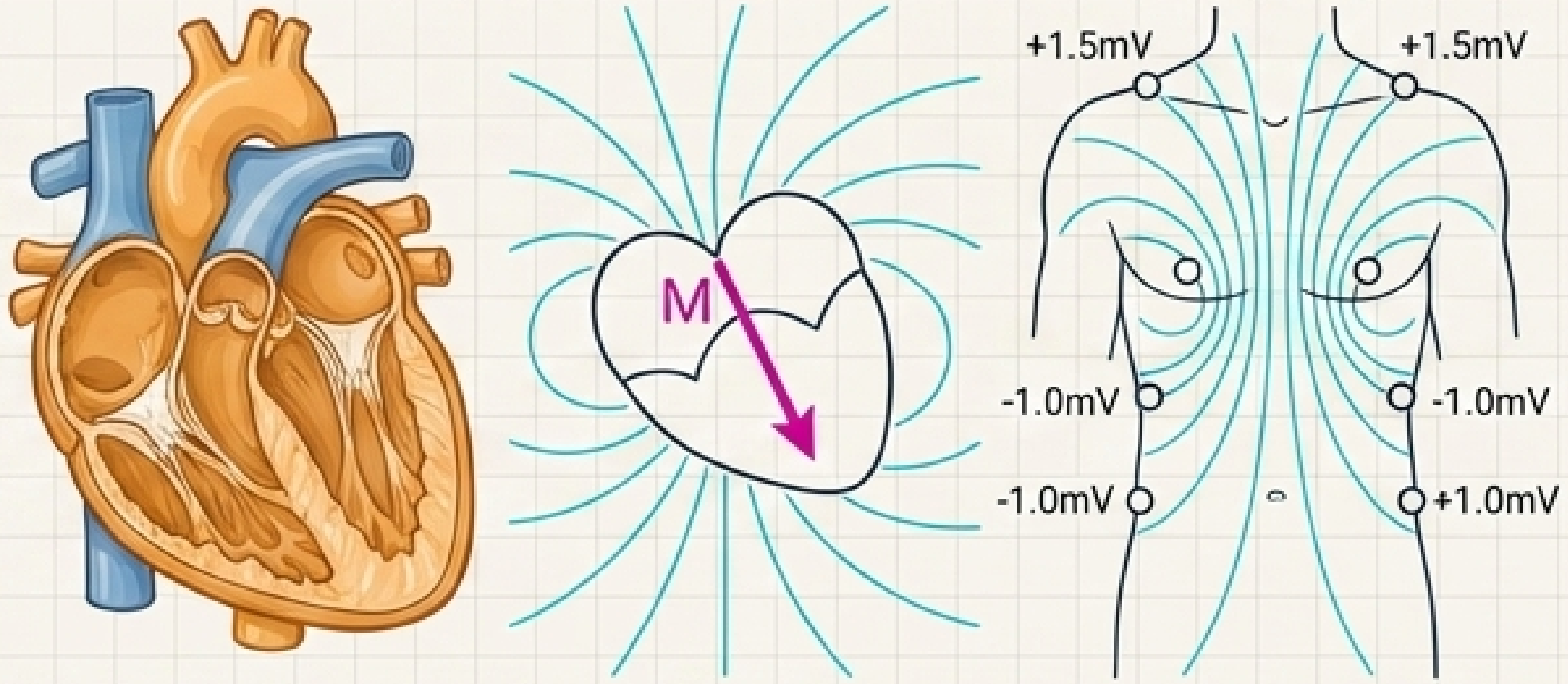
Application: The Biological Dipole

The Ideal Dipole



$$\text{Field Strength Proportional to } \frac{1}{r^3}$$

The Biological Engine



The depolarization of heart muscle cells forms a changing **dipole moment**. The resulting electric field extends to the skin surface, generating the **precise voltage differentials** measured by an **Electrocardiogram** (ECG).

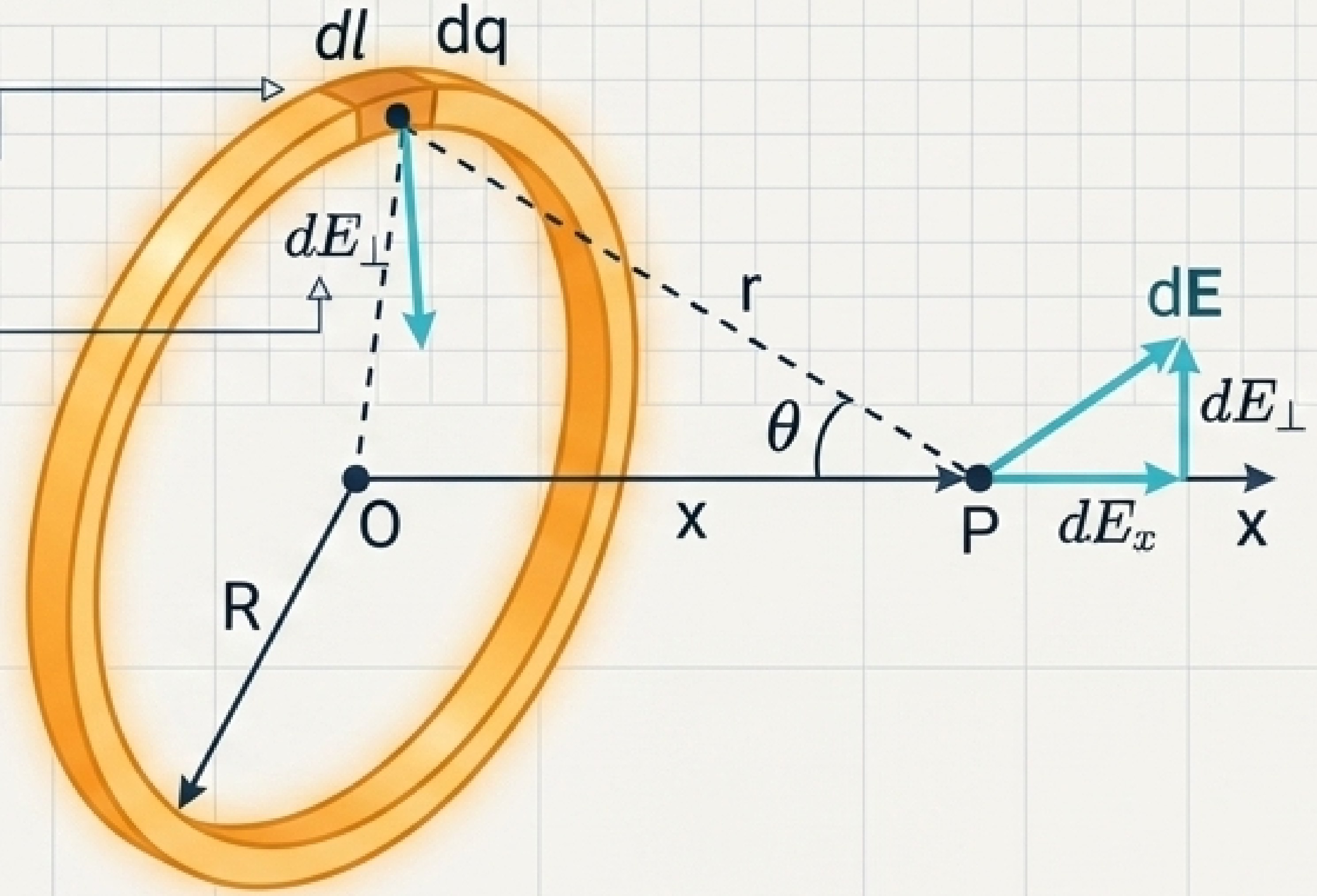
Blueprinting the Field: The Charged Ring

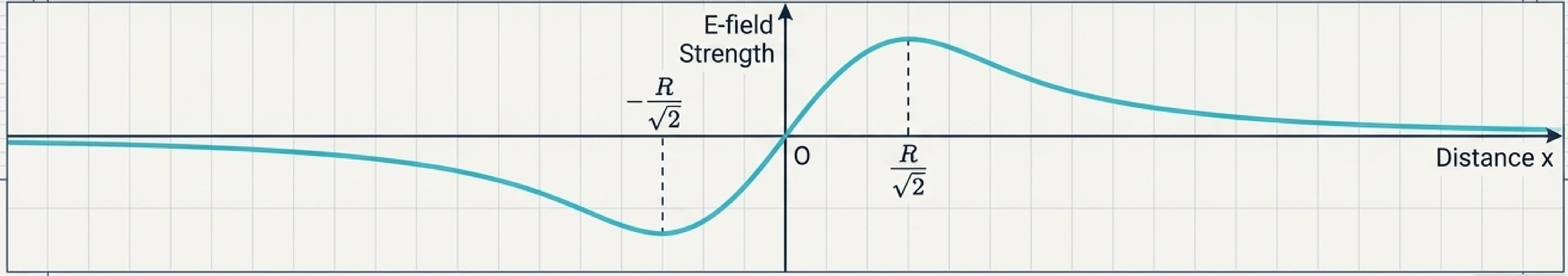
1. Define the charge segment: $dq = \lambda dl$

2. Leverage geometric symmetry:
$$\int dE_{\perp} = 0$$ (Perpendicular forces cancel)

3. Integrate along the axis to solve.

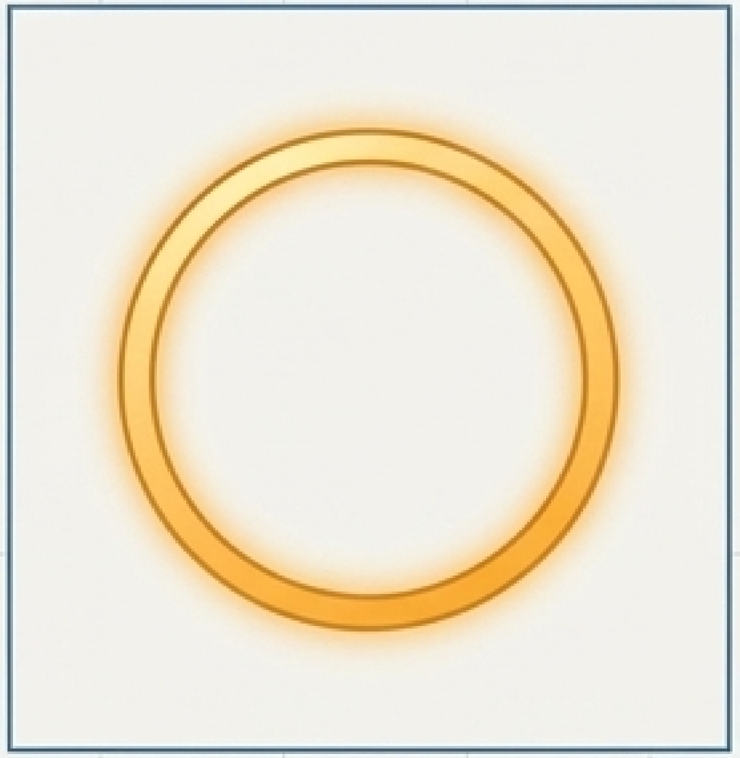
$$E = \frac{qx}{4\pi\varepsilon_0 (x^2 + R^2)^{3/2}}$$



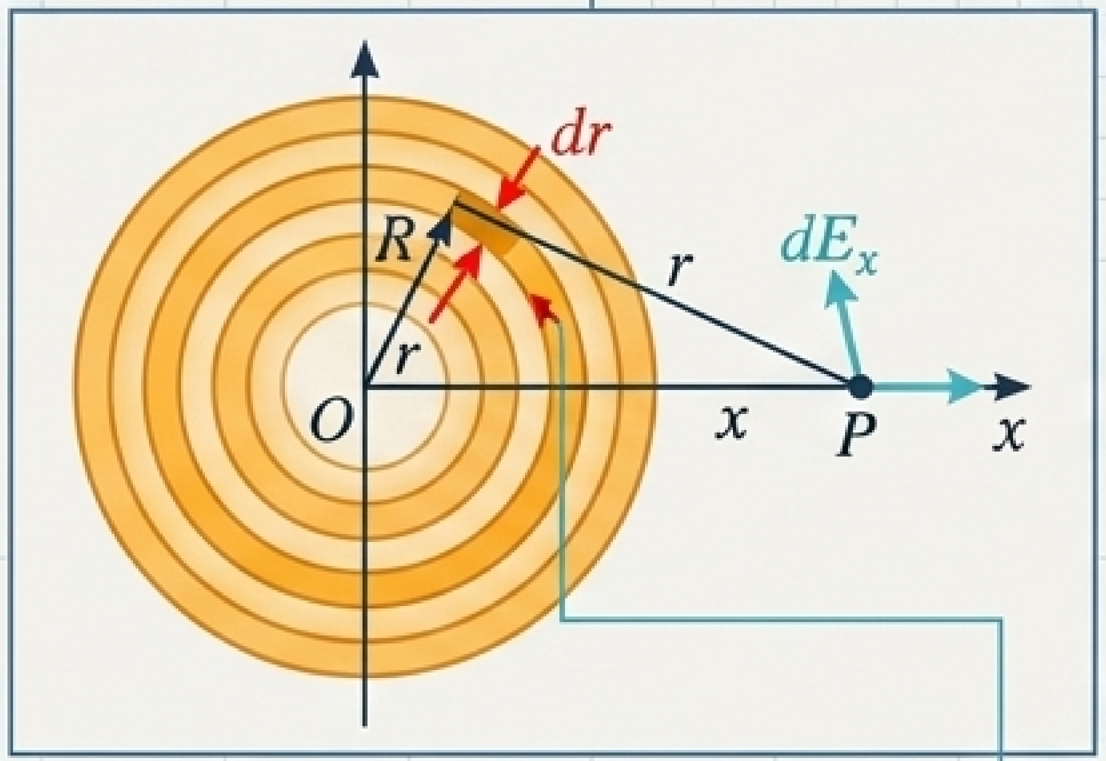


Dimensional Scaling: From Ring to Infinite Plane

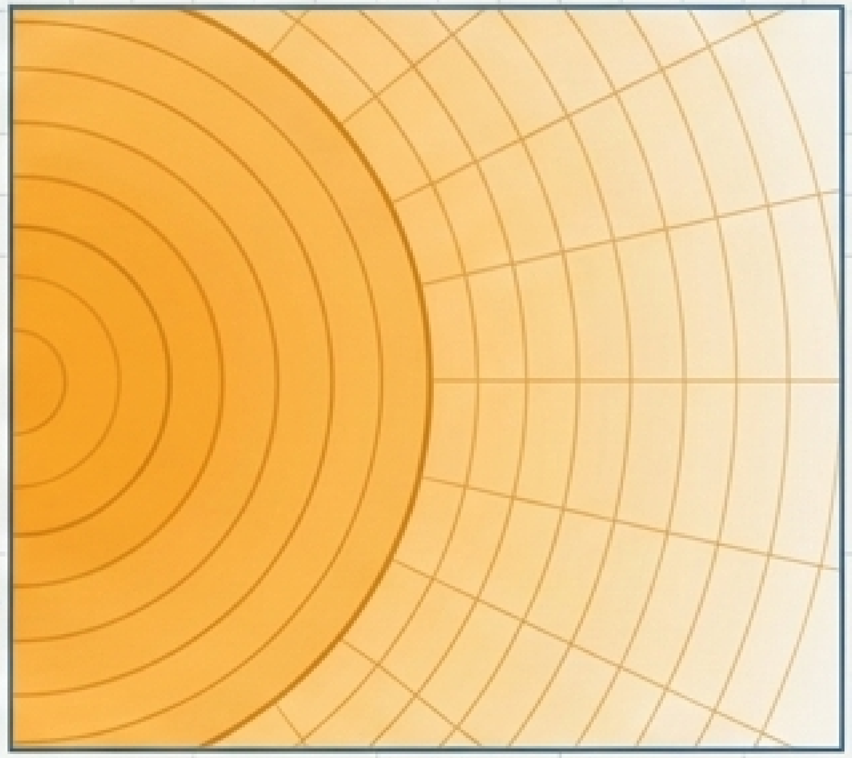
The 1D Ring



The 2D Disk



The Infinite Plane



Integrate concentric rings:

$$dq = \sigma 2\pi r dr$$

$$dE_x = \frac{\sigma x r dr}{2\varepsilon_0 (x^2 + r^2)^{3/2}}$$

$$E = \int dE_x = \frac{\sigma x}{2\varepsilon_0}\left(\frac{1}{\sqrt{x^2}} - \frac{1}{\sqrt{x^2 + R^2}}\right)$$

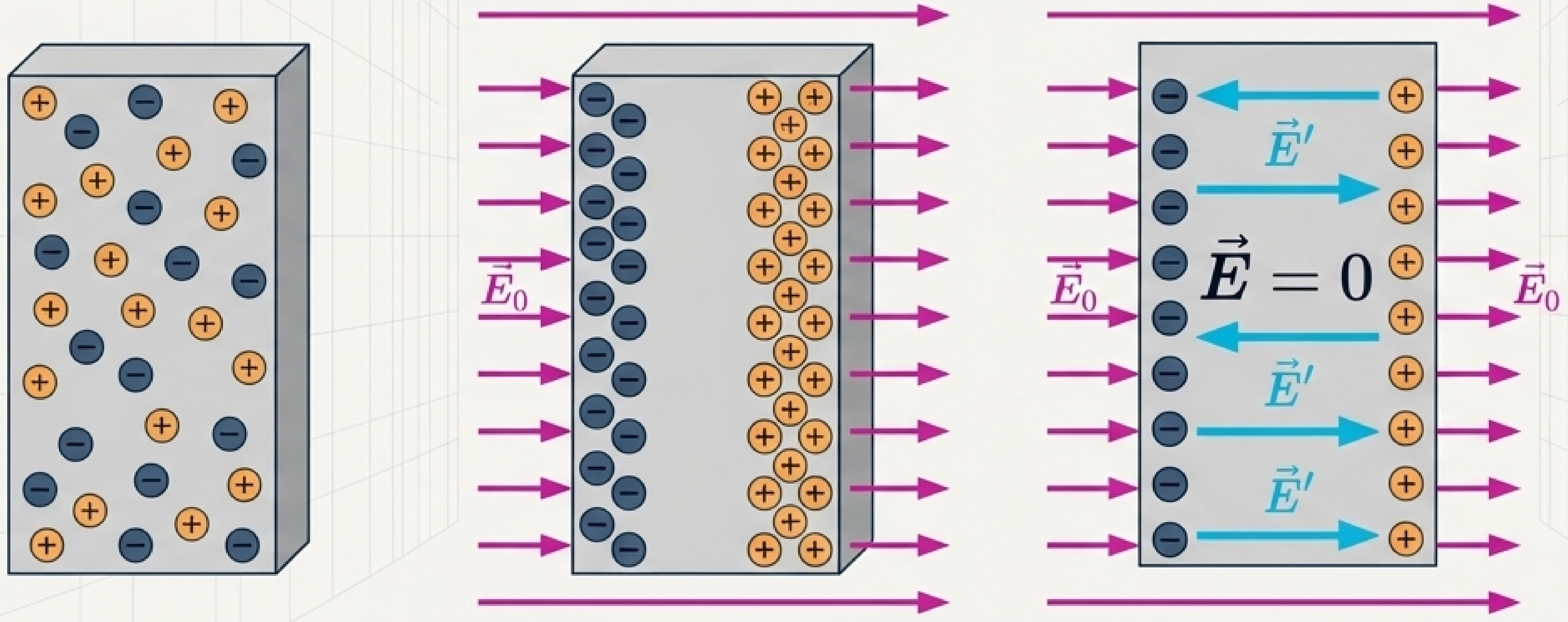
$$x \ll R \implies \sqrt{x^2 + R^2} \approx R$$

$$E \approx \frac{\sigma}{2\varepsilon_0}\left(1 - \frac{x}{R}\right) \approx \frac{\sigma}{2\varepsilon_0}$$

Infinity is a practical concept: any flat charged surface mathematically behaves as an 'infinite plane' if the observation point is close enough to the surface.

Electrostatic Equilibrium: The State Machine

Phase 1: Neutrality → **Phase 2: Induction** → **Phase 3: Equilibrium**



Phase 1: Uncharged solid conductor, charges randomized.

Phase 2: External field causes charge separation.

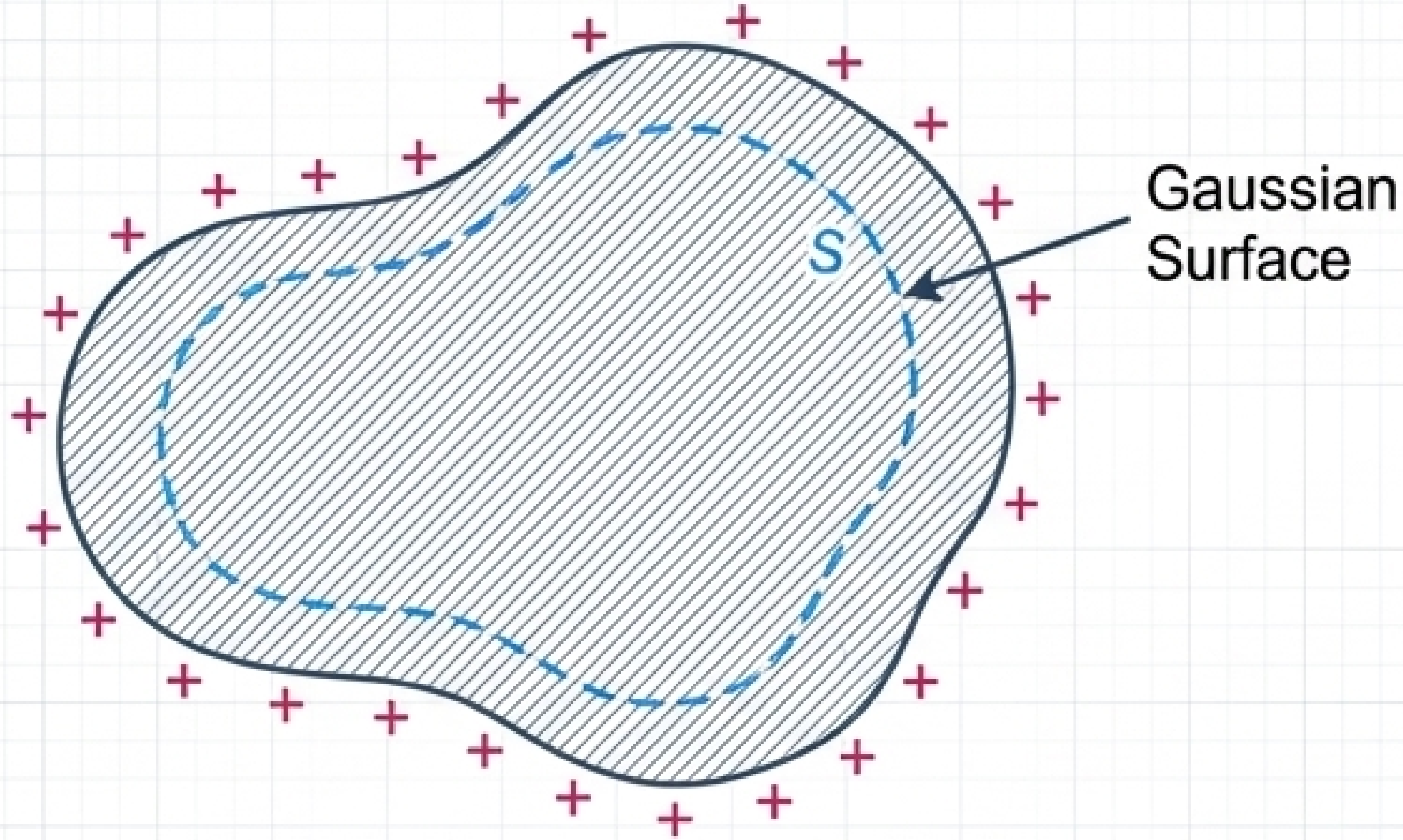
Phase 3: Internal counter-field cancels external field, net $\vec{E}$ is zero.

The 4 Golden Rules of Conductors

1. Inside the conductor, $\vec{E} = 0$.
2. The entire conductor acts as an equipotential volume.
3. Moving charge along the surface requires zero work.
4. External electric field lines must intersect the surface perpendicularly.

Conductor Topology: Solid vs. Hollow Dynamics

Solid Conductors



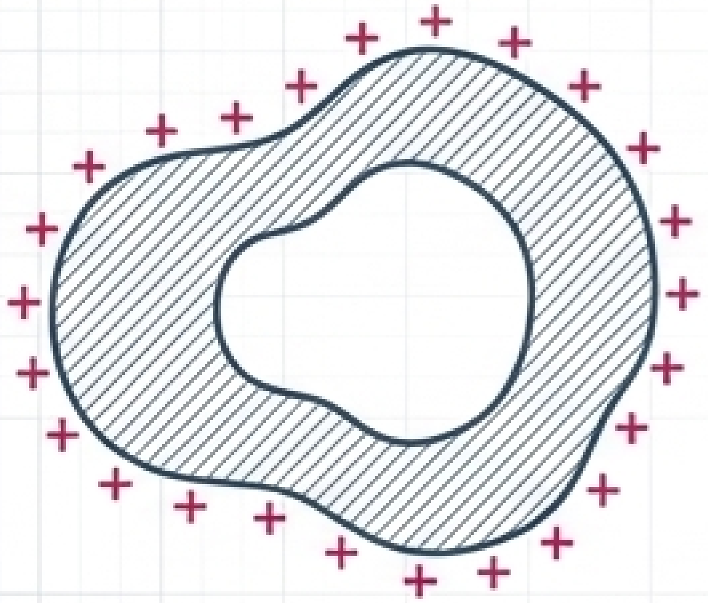
$$\oint_S \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0} \Rightarrow$$

Since $\vec{E} = 0$ inside conductor, $q_i = 0$

All charge *q* resides on the outer surface

Conclusion: Inside a conductor, there is no net charge; charge resides only on the conductor's outer surface.

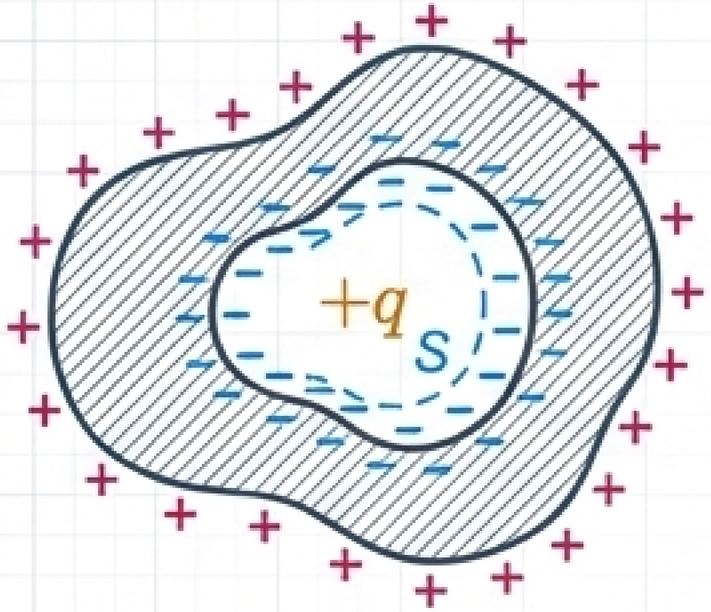
Hollow Cavity Exceptions



Scenario A: Empty Cavity

$$\oint_S \vec{E} \cdot d\vec{S} = 0 \Rightarrow q_i = 0$$

If there is no charge inside the cavity, charge distribution is only on on the outer surface, and the inner surface is neutral.



Scenario B: Charge Inside Cavity

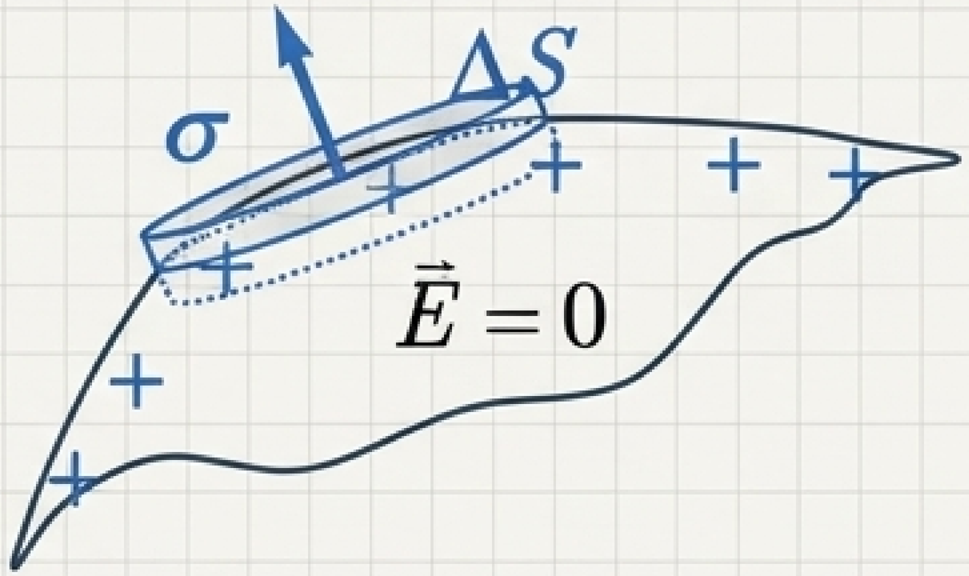
$$\oint_S \vec{E} \cdot d\vec{S} = \sum_i q_i = 0$$

- Inner surface has an induced charge $-q$
- Outer surface has an induced charge $+q$

Induction Chain: The $+q$ charge in the cavity induces a net $-q$ charge on the inner surface to cancel the field inside the conductor, which forces an equal and opposite $+q$ charge to the outer surface to maintain neutrality.

Surface Curvature & Point Discharge

The Curvature Rule

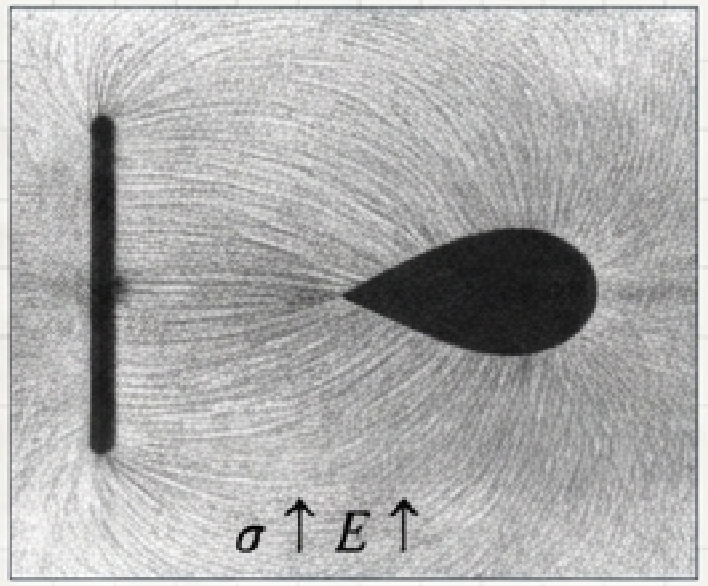


$$\oint_S \vec{E} \cdot d\vec{S} = E\Delta S = \frac{\sigma \Delta S}{\epsilon_0}, \quad E = \frac{\sigma}{\epsilon_0}$$

$$\sigma \uparrow E \uparrow ; \sigma \downarrow E \downarrow$$

The **smaller** the radius of curvature (R), the higher the surface charge density (σ). Extreme sharp points hoard charge.

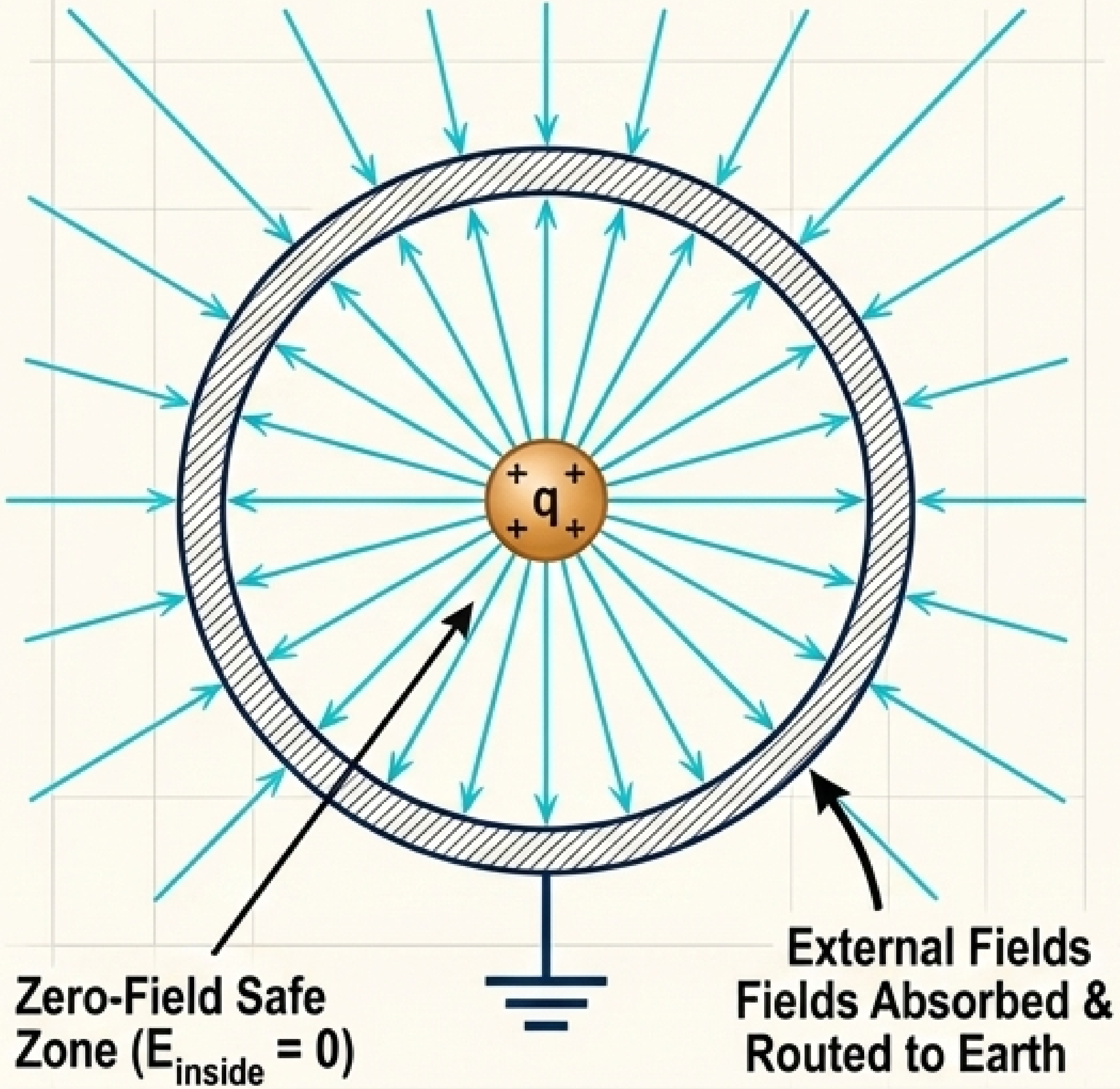
Application: The Lightning Rod



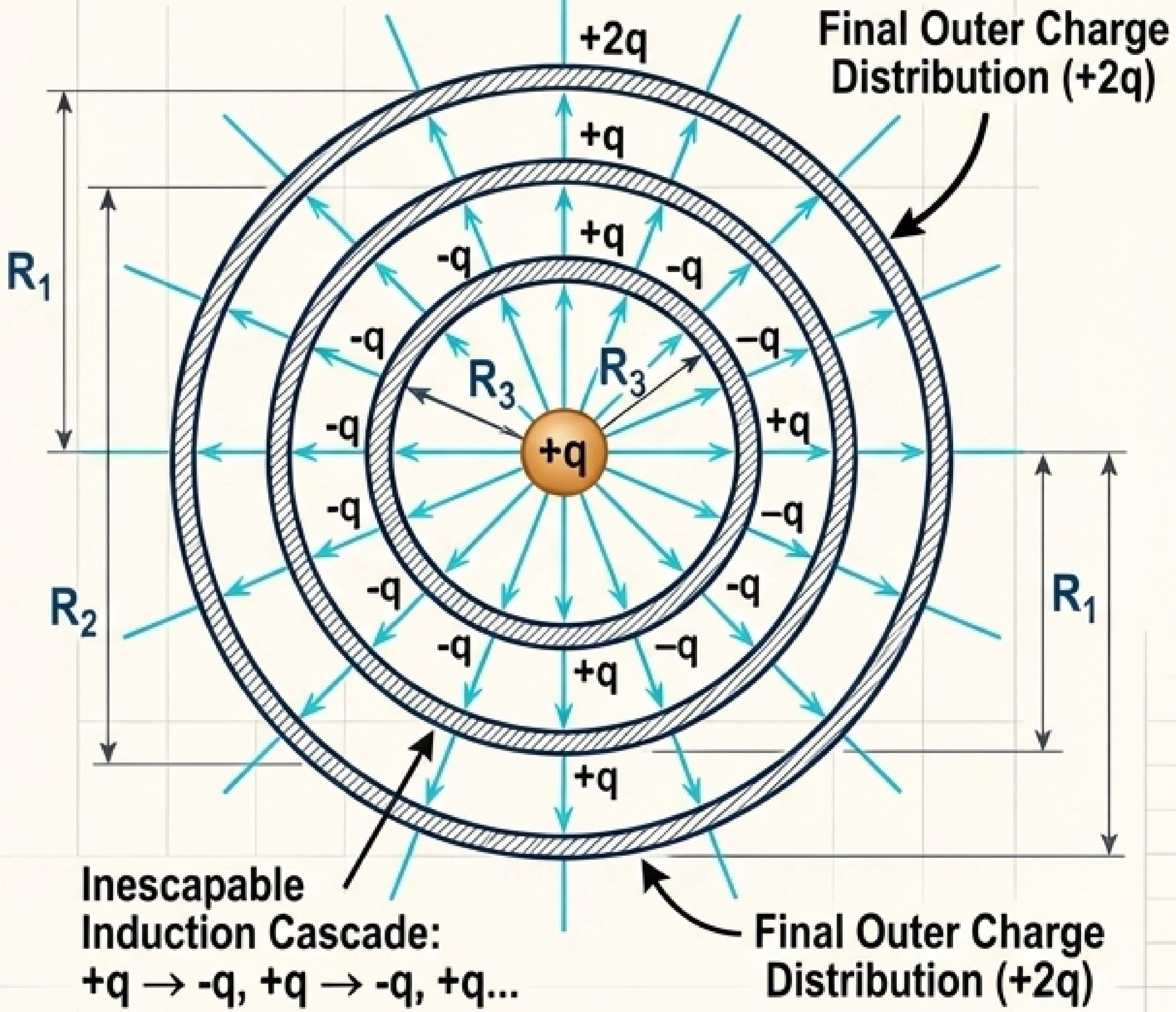
The extreme E-field at the sharp tip ionizes the surrounding air, creating a conductive path to safely neutralize the cloud's charge to the Earth.

Multi-Layer Routing & The Faraday Cage

Electrostatic Shielding (The Absolute Void)



Complex Concentric Routing



NotebookLM

Defining Capacitance: The Engineering of Empty Space

Capacitance Calculation Framework

(1) Assume $\pm Q$.

(2) Find $\vec{E}$.

(3) Find Potential U.

(4) Solve $C = Q/U$.

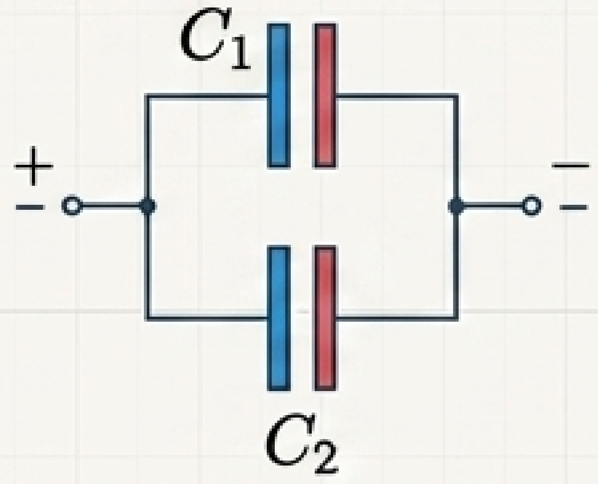
Parallel Plate Application

$$C = \frac{\varepsilon_0 S}{d}$$

C depends solely on geometry and dielectric, **never** on Q.

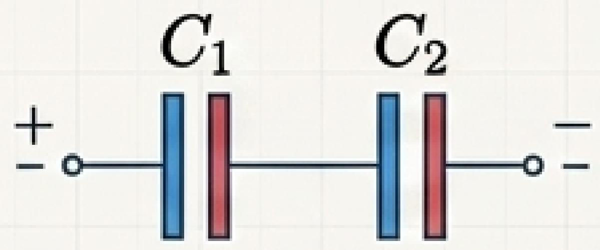
Circuit Topologies: Series & Parallel

Parallel Connection



$$C = C_1 + C_2$$

Series Connection



$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

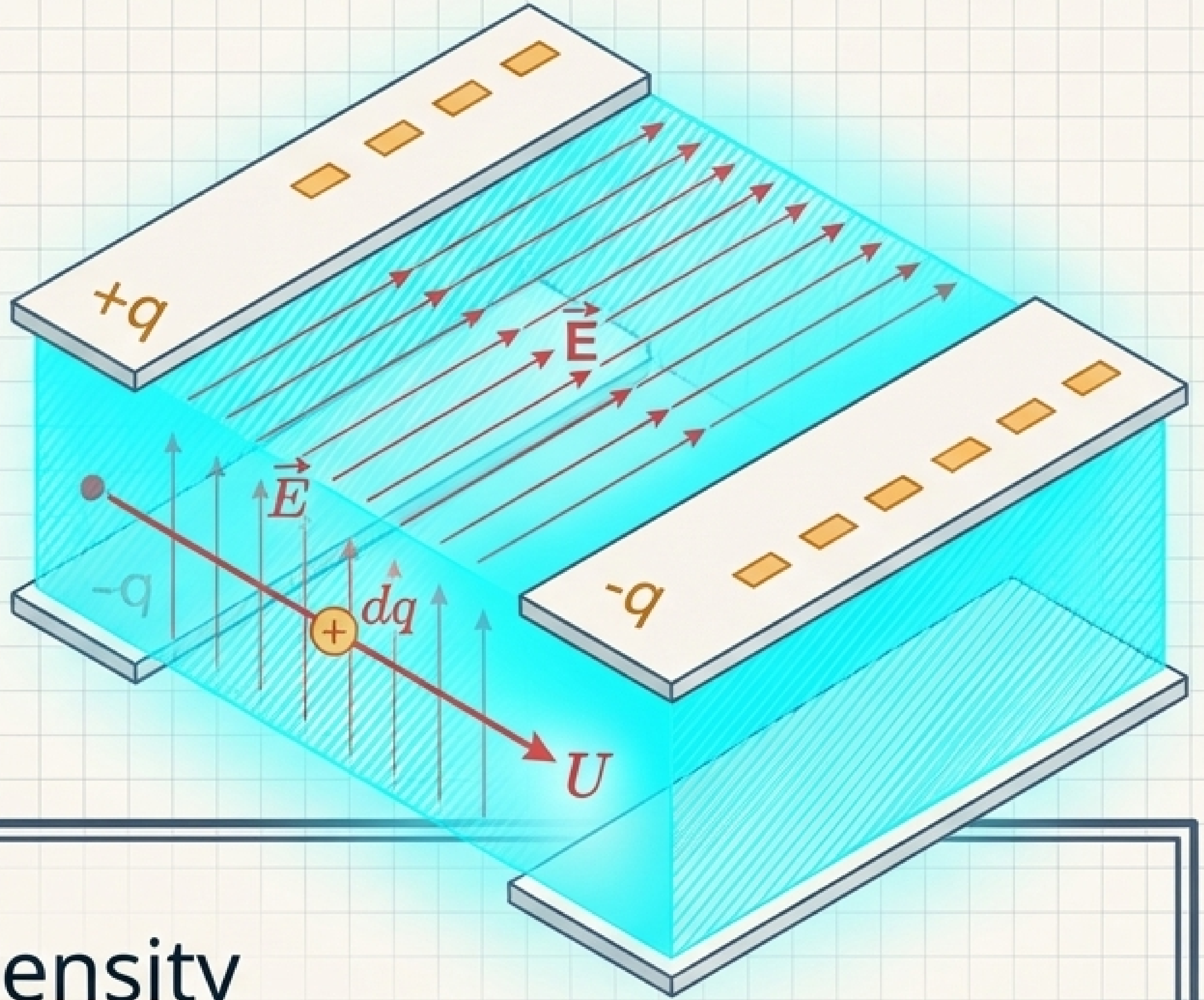
Energy Density: Storing Power in the Void

$$dW = U\,dq = \frac{q}{C}\,dq$$

$$W = \frac{1}{C}\int_0^Q q\,dq = \frac{Q^2}{2C}$$

$C = Q/U$

$$W_e = \frac{Q^2}{2C} = \frac{1}{2}QU = \frac{1}{2}CU^2$$



$$\boxed{w_e = \frac{1}{2}\varepsilon_0 E^2 \text{ Energy Density}}$$

Energy does not live on the metal plates; the plates merely anchor the field.
The energy is physically embedded in the cubic volume of the field itself.

Synthesis: The Tapestry of Physics

The Relativistic Universe

$$E = mc^2$$

Energy has inertia.
Highly concentrated energy manifests as physical mass.

The Electrostatic World

$$w_e = \frac{1}{2}\varepsilon_0 E^2$$

Empty space has structure.
Invisible fields physically hold energy in the void.

Whether examining the crushing speeds of relativity or the silent tension of an electric field, modern physics reveals one unified truth:

Mass and **Empty Space** are not passive stages for events to happen—they are active, malleable structures woven entirely out of **Energy**.